



# Your Section Title

An optional subtitle or tagline



Lectures



proof.lean



README



Exercises





# LEAN: Proof Assistant Demo Template

- Structural Example: Definition
- Key Axioms & Lemmas
- Theorem Application Steps
- Formalization Discussion

```
Lean Editor.pro
Monaco
1 theorem add_assoc (n m k : N) : n + m + k = n + (m + k)
  :=
2 begin
3   induction n with n' hn,
4     rw add_zero, -- Base Case: 0 + m + k = 0 + (m + k)
5     rw zero_add,
6     rw add_succ, -- Inductive Case: succ n' + m + k =
7                   succ n' + (m + k)
8     rw hn,       -- Use inductive hypothesis
9     rw add_succ, -- etc.
10  end
```



**TIP:** Press CMD+P to Verify.

?



# Your Slide Title Here

- ❑ Your first point
- ❑ Your second point
- ❑ Your third point
- ❑ Your fourth point

☐ filename.ext ☐

Monaco

```
1  -- Replace this with your own code, pseudocode, or text.  
2  -- The box wraps long lines automatically.  
3  your_function(argument):  
4      return result
```



**TIP:** A short tip or callout goes here.

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# LEAN: Complete Proof Walkthrough

```
Lean Editor.pro
Monaco

1 theorem mul_comm (n m : N) : n * m = m * n :=
2 begin
3   induction n with n' hn,
4     rw zero_mul,      -- Base Case: 0 * m = m * 0
5     rw mul_zero,
6     rw succ_mul,      -- Inductive Case: succ n' * m = m * succ n'
7     rw hn,
8     rw mul_succ,
9 end
```



**TIP:** Press CMD+P to Verify.

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# LEAN: Course Roadmap

- Week 1: Propositional Logic
- Week 2: Natural Numbers & Induction
- Week 3: Lists & Recursion
- Week 4: Tactics Deep Dive
- Week 5: Building a Library
- Week 6: Final Project



**TIP:** No code today - just planning.

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# LEAN: Simplifying a Proof

Before

```
□ proof_v1.lean ✕  
Monaco  
1 theorem add_zero'  
2   (n : N) : n + 0 = n :=  
3 begin  
4   induction n with n' hn,  
5     refl,  
6     rw add_succ,  
7     rw hn,  
8 end
```

After

```
□ proof_v2.lean ✕  
Monaco  
1 theorem add_zero'  
2   (n : N) : n + 0 = n :=  
3 by induction n;  
4   simp [*, add_succ]
```

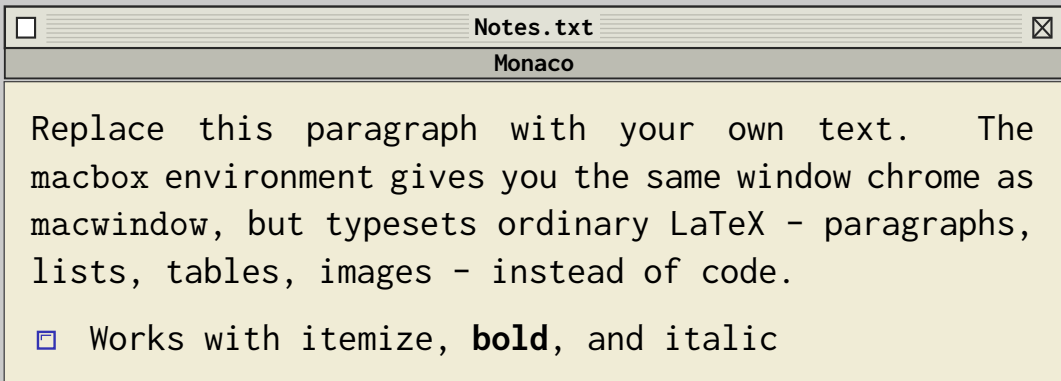


**TIP:** The simp tactic closes both cases at once.

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# A Note, Definition, or Quote



**TIP:** Swap this box for a table, image, or blockquote.

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# LEAN: Simplifying a Proof

```
proof_v1.lean
Monaco

1 theorem add_zero'
2   (n : N) : n + 0 = n :=
3 begin
4   induction n with n' hn,
5     refl,
6     rw add_succ,
7     rw hn,
8 end
```



**TIP:** The simp tactic closes both cases at once.

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## By the Numbers

128

theorems proved

42

lemmas reused

6

weeks of work



**TIP:** Swap in your own three (or more) headline numbers.

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